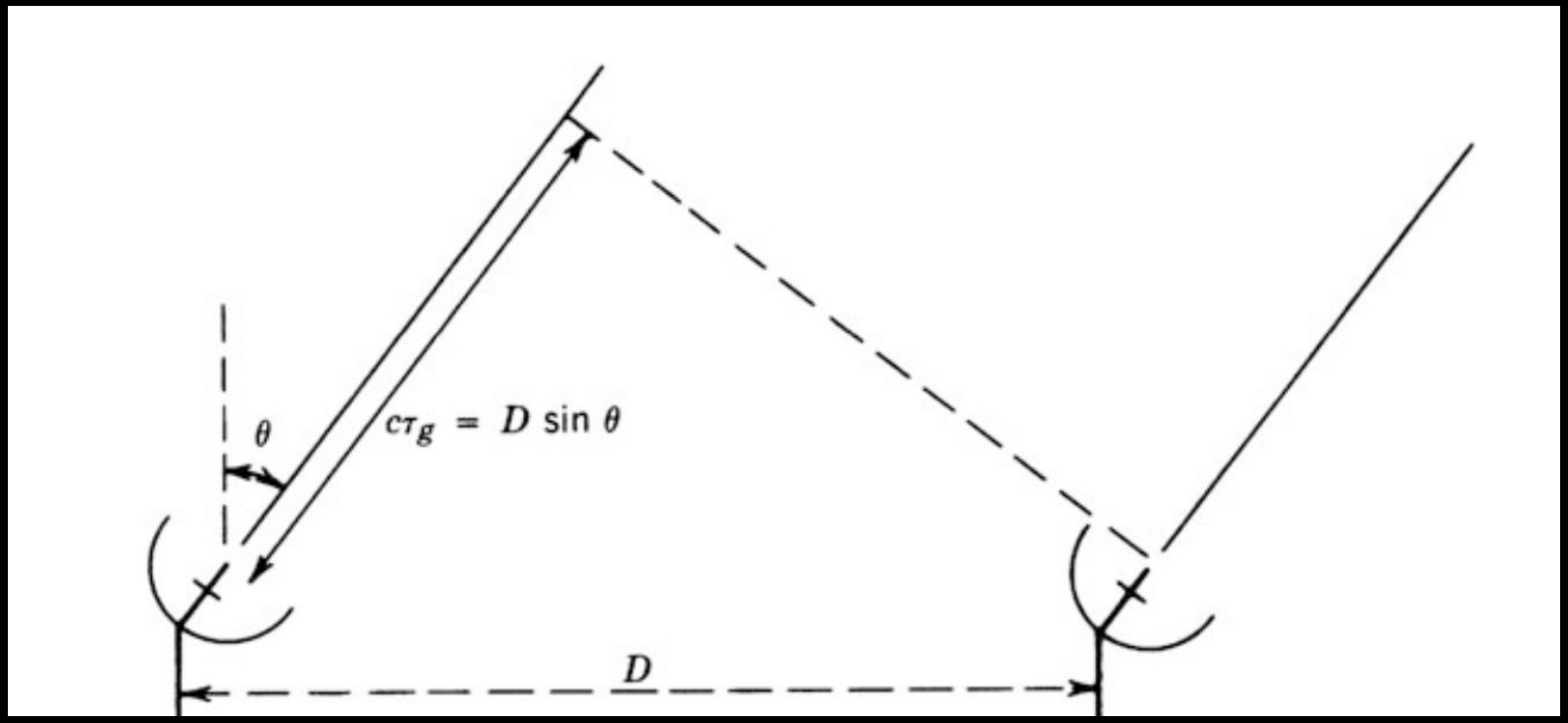
Two Element Interferometer

G N Paneendra
Radio Astronomy Winter School 2026

21 - 02 - 2026

Two-Element Interferometer team:
Abhishek R, Arul Pandian B, Arutkeerthi, Deep Bhowmik, Janardhana, Katta Naveen, Krishna Balaji, Likhit, Prabu T, Praveen B V, Raghavendra Rao K B, Roshan Dora, Sahana B, Srinath, Vinod K R

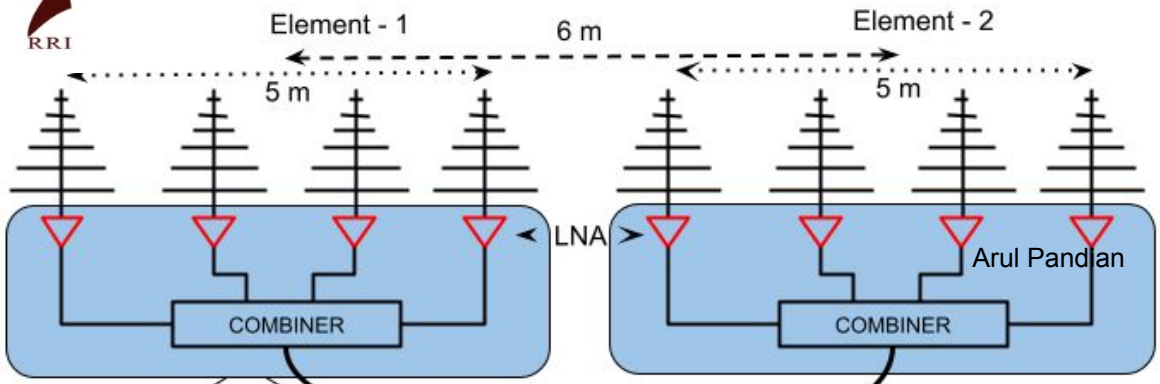
Interferometer



Two-Element Radio Interferometer

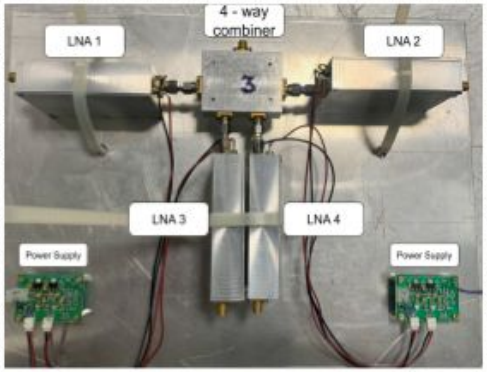
Location: Gauribidanur Radio Observatory
Latitude: 13.60° N; Longitude: 77.44° E
Baseline: 8.5m
Operating Frequency: 180 - 350 MHz

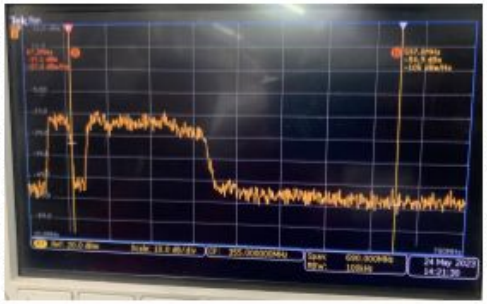
Two - Element Interferometer RF Signal Path



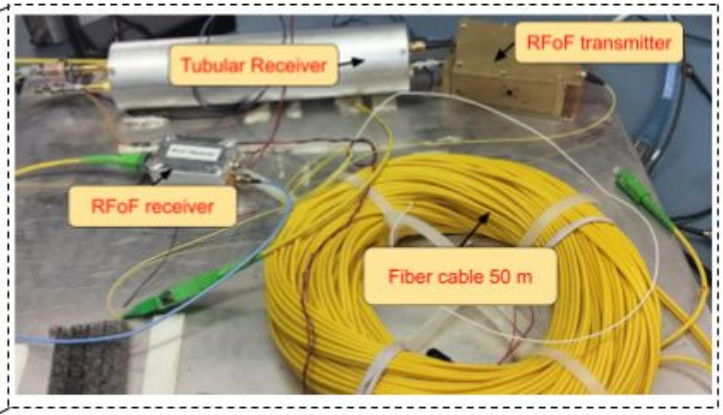
Arul Pandian

Picture Credit: Abhishek R,





LNA response with notch filter



Tubular receiver

Designed by K B Raghavendra Rao

RFoF Transmitter x 2

Design by Abhishek R, Keerthipriya, Somashekar

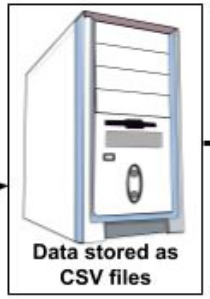
50 m Fiber cable

RFoF Receiver x 2

Element - 1

Element - 2

Oscilloscope



Ethernet

Intranet

Remote systems n

Remote system 1

Abhishek R, Arul Pandian B, Arutkeerthi, Deep Bhowmik, Janardhana, Katta Naveen, Krishna Balaji, Likhit, Prabu T, Raghavendra Rao K B, Roshan Dora, Sahana B, Srinath, Vinod K R

Observation Scheduling

Streamlit

Not secure 172.17.20.227:8501

Finish update

Stop

Navigation

Choose the task you want to do

Data Acquisition

Calculator

Frequency Bin Calculator

Developers:

A.A.S.LIKHIT

astropi.2003@gmail.com

Katta Naveen

naveenkatta626@gmail.com

LPDA Data Acquisition

Enter your name:

VINOD K R

Enter the IP address of the oscilloscope:

172.17.20.55

Enter the active channels separated by comma (e.g., CH1,CH3):

CH1,CH2,CH3,CH4

Enter the start time (in format YYYY-MM-DD HH-MM):

2025-11-26 08-49

Enter the end time (in format YYYY-MM-DD HH-MM):

2025-11-26 17-00

Enter the delay time in seconds between each data collection(min 2 seconds):

3.00 − +

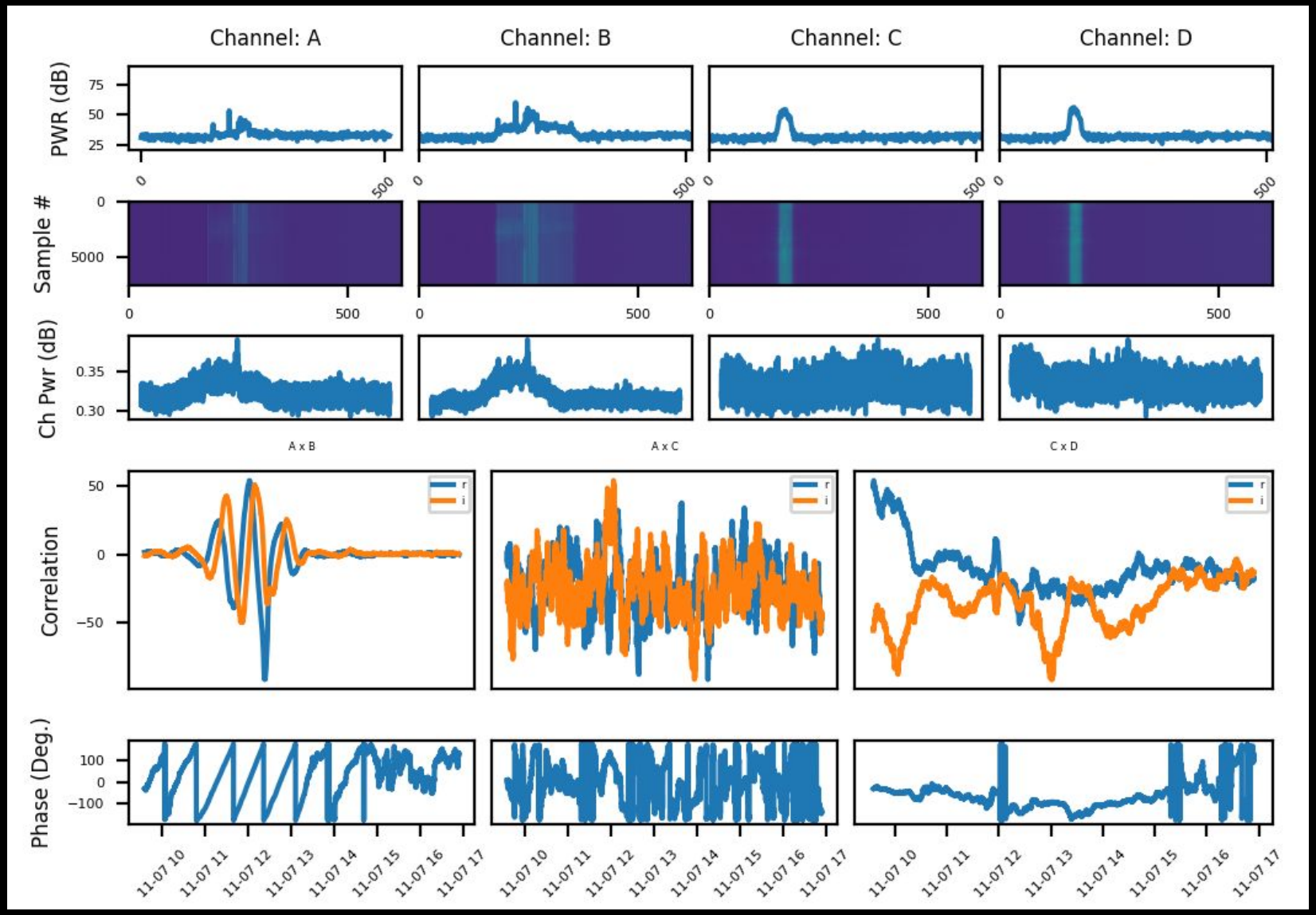
Enter the name of the folder to save the files:

26Nov25SUN

Start data acquisition

Credits: Likith and Naveen

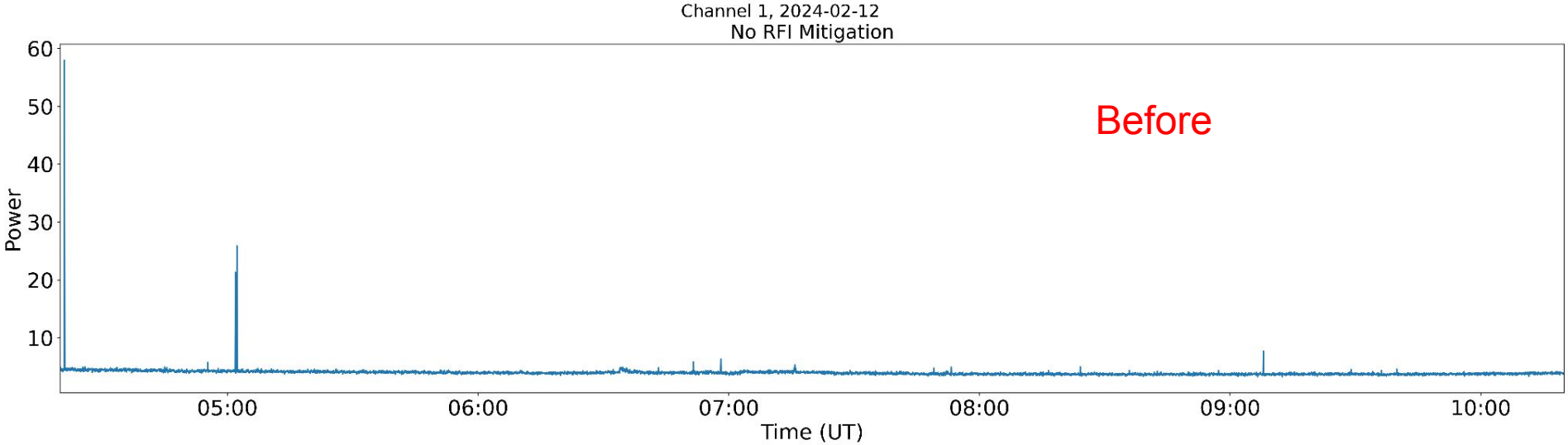
Real-Time Monitoring

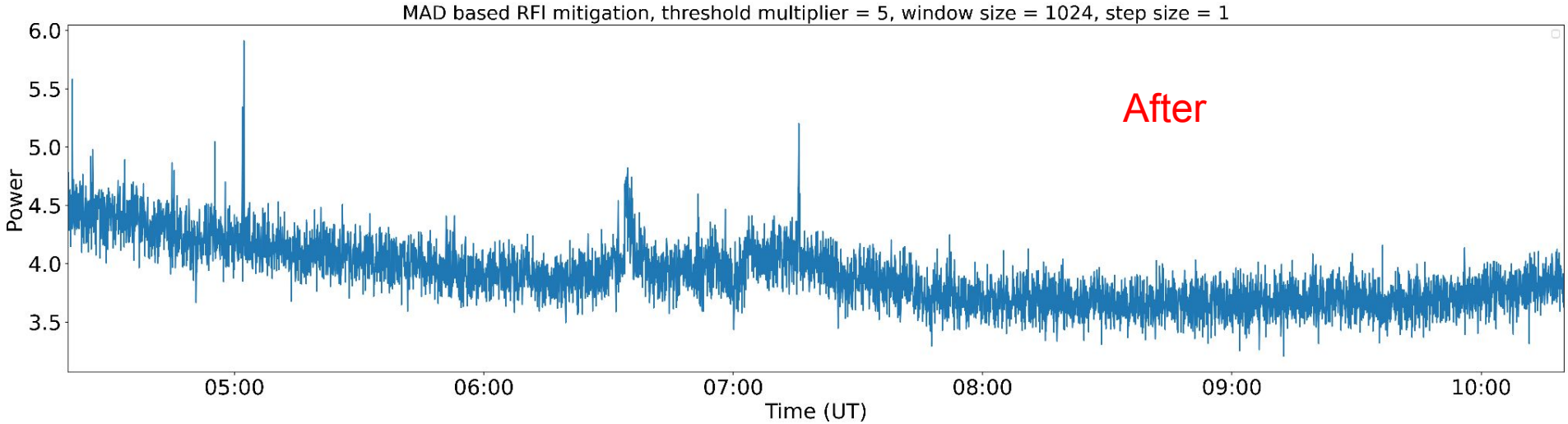


Credit: Arul Pandian B

Data Processing

Results of MAD based RFI Mitigation (Time Domain)





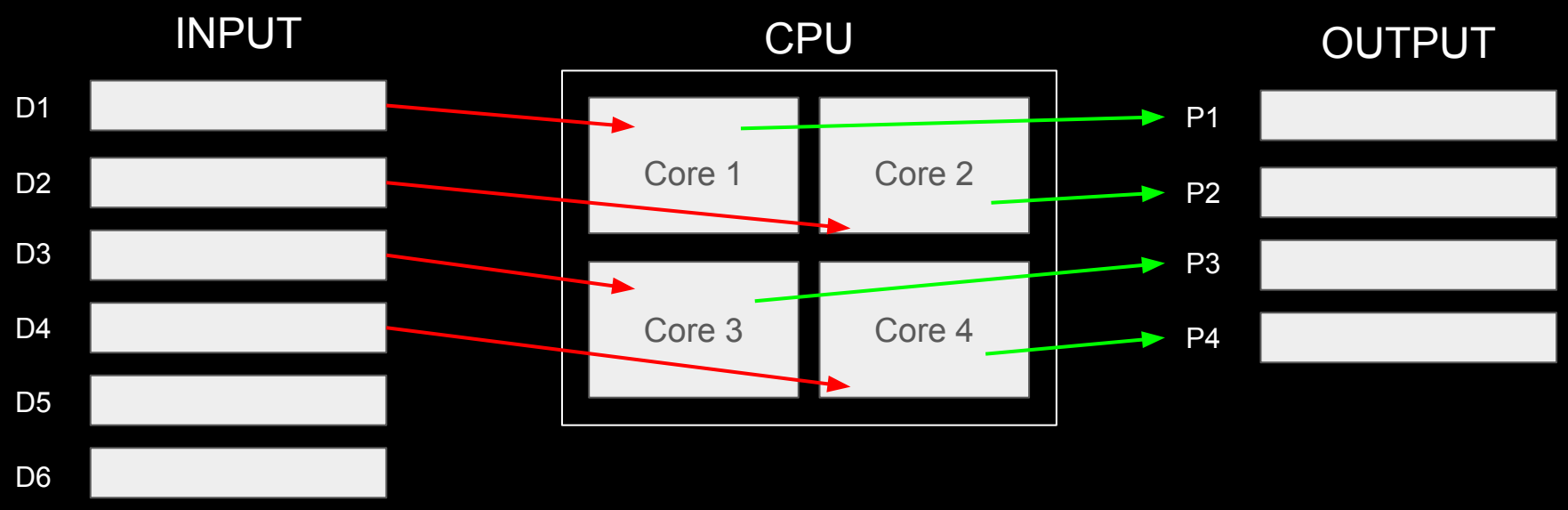
Parallelisation of RFI Mitigation algorithm on RRI HPC

Before

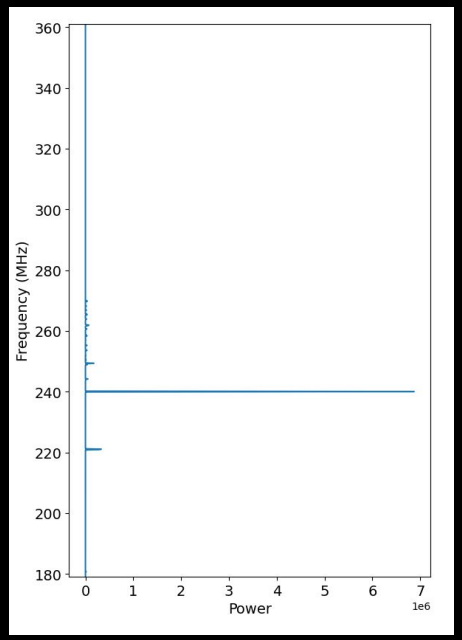
- Time taken = ~55 mins
- Process runs through single core

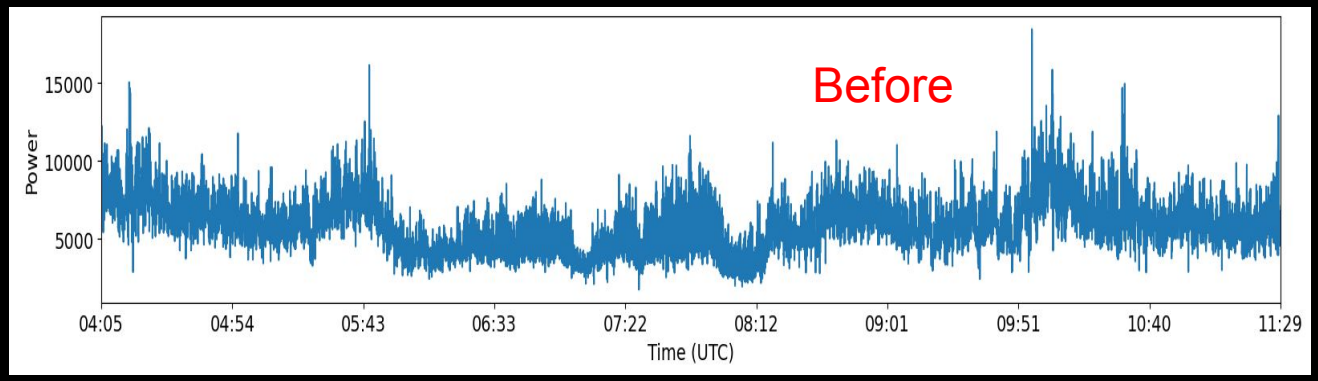
After

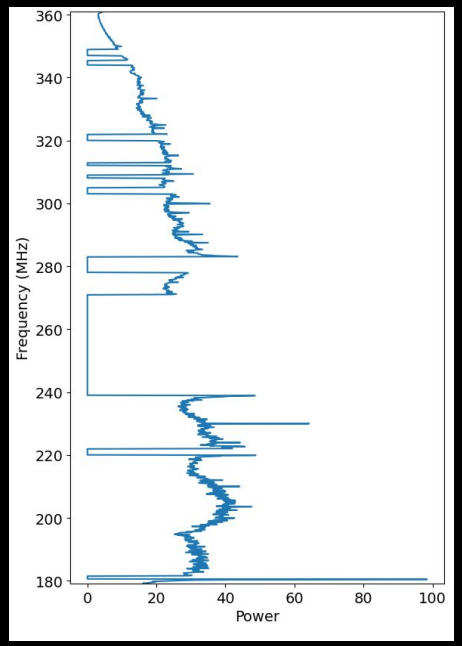
- Time taken = ~5 mins
- Process runs through 32 cores

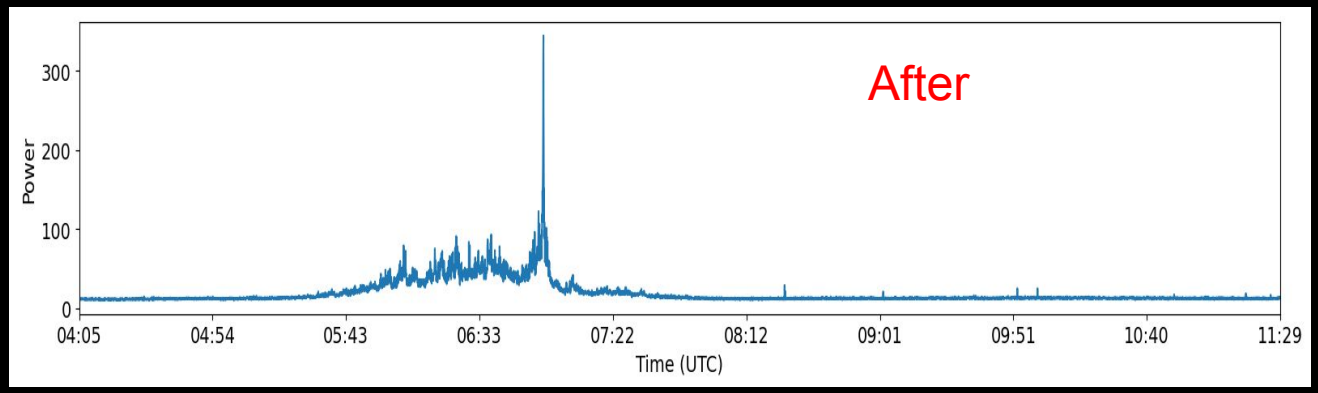


Results of RFI Mitigation (Frequency Domain)

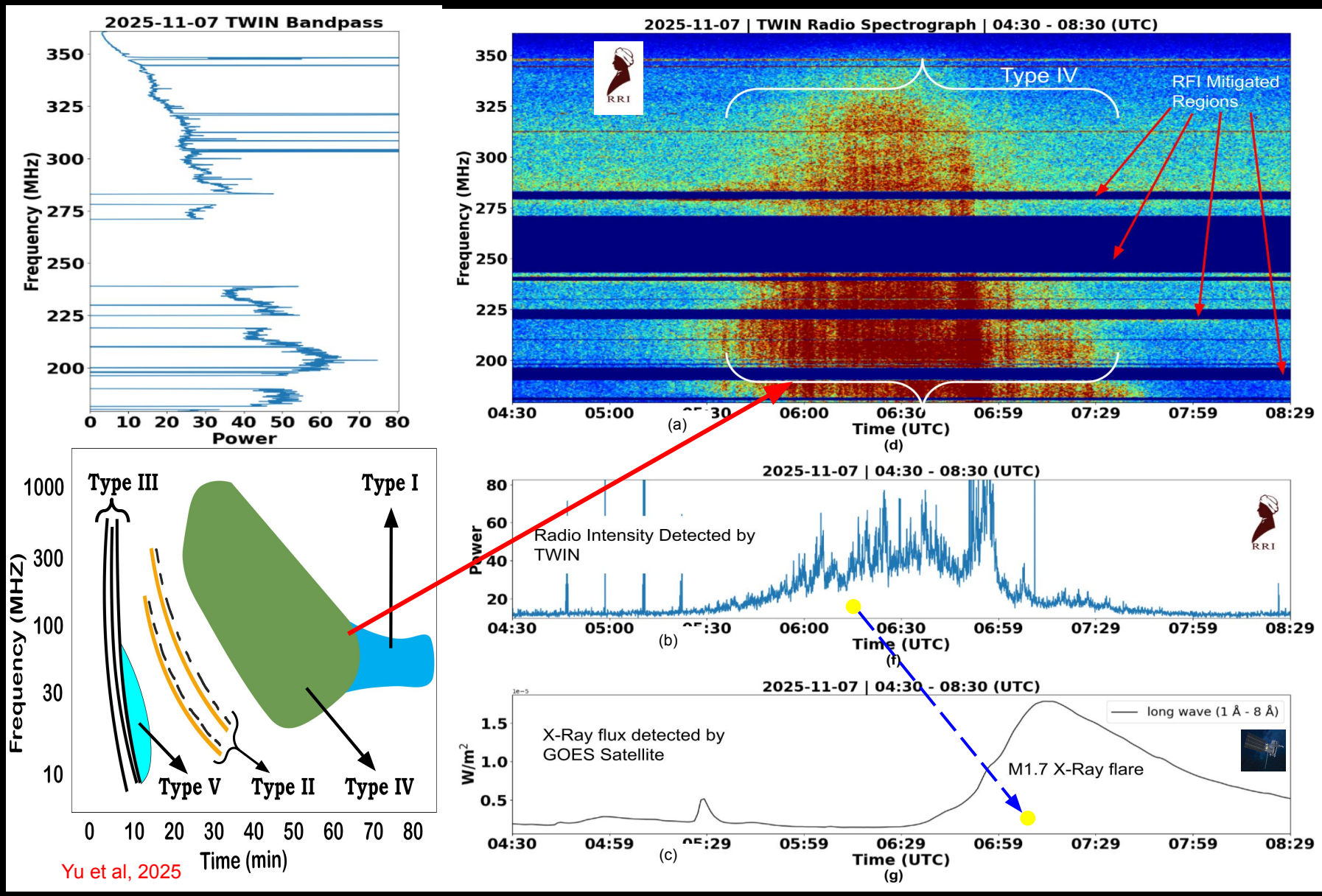






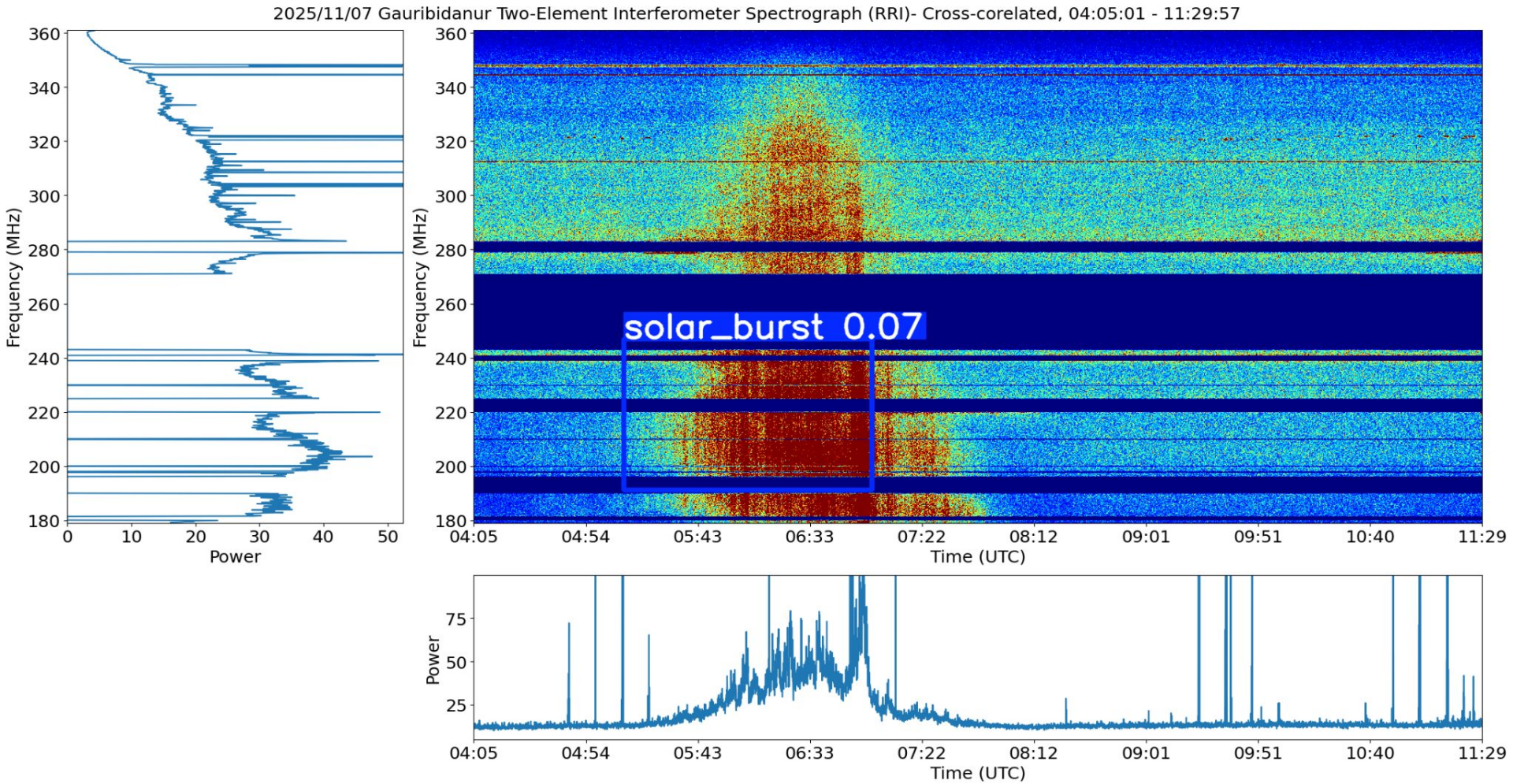


A Major Solar Storm detected by RRI G-TWIN ahead of satellite detection

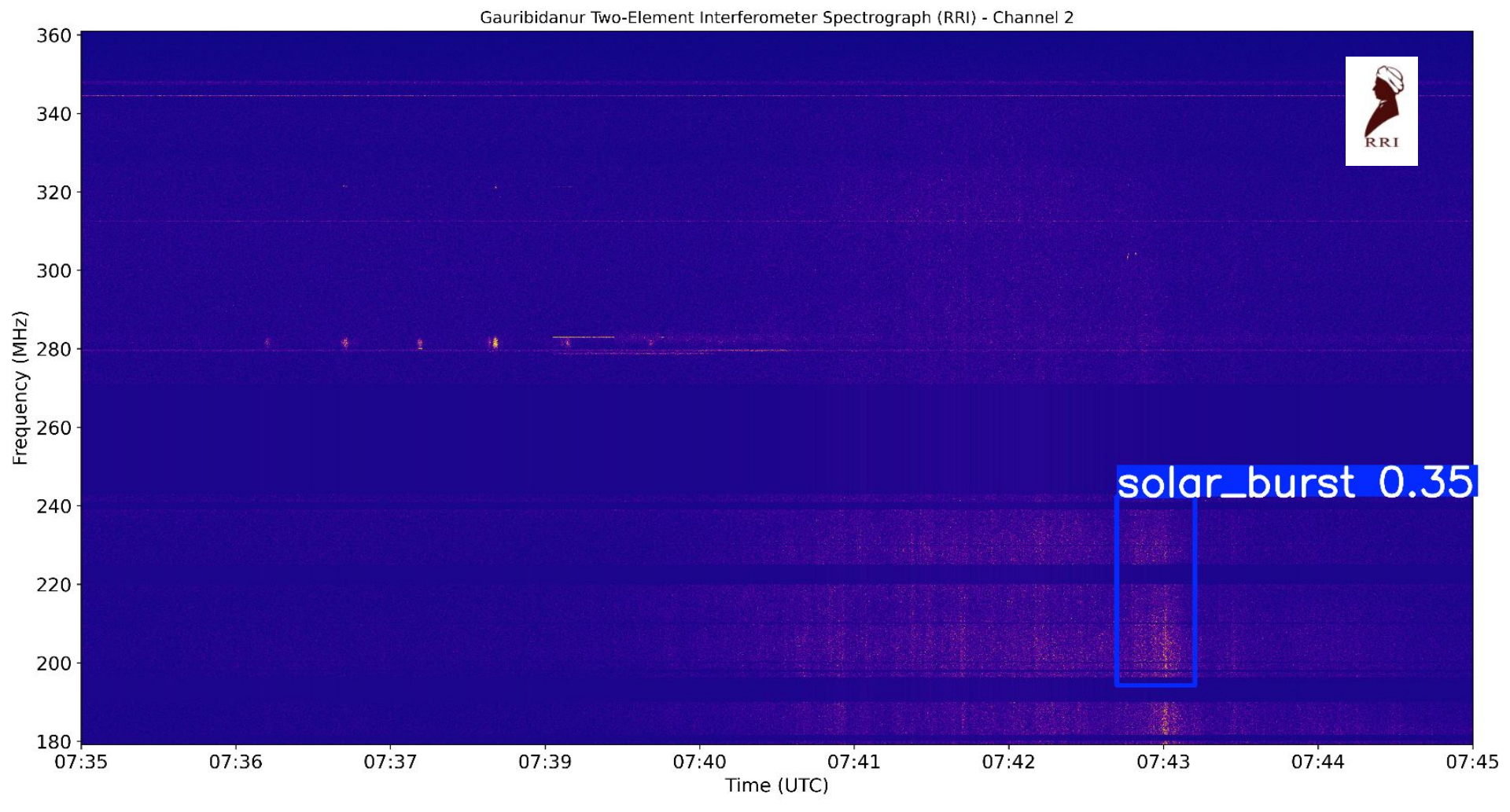


Yu et al, 2025

ML based automated Solar Burst Detection using YOLO11



ML based REAL-TIME automated Solar Burst Detection using YOLO11



TWIN Solar Transit Data

Gauribidanur Two-Element Interferometer

Location: Gauribidanur Radio Observatory, Raman Research Institute
Latitude: 13.60° N; Longitude: 77.44° E

Webpage developed to access the data 🎀

Instrument Specifications

- Frequency bandwidth = 180 MHz - 360 MHz
- Sampling rate (ADC rate) = 1.25GS/s
- ADC resolution = 8bit
- Baseline = 6 m

Additional Downloads

README.pdf | twin_data_process_raw.py | twin_data_process_combined.py

2023 | 2024 | 2025

January | February | March | April | May | June | July | August | September | October | November | December

2024, February

Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29		

IP address
172.16.101.87:5001

Search the IP in a web browser to find the webpage!!!!

<u>Reference:</u>

1. The Scientist and Engineer's Guide to Digital Signal by Steven W Smith
2. Interferometry and Synthesis in Radio Astronomy by Richard Thompson
3. Antennas by John D Kraus